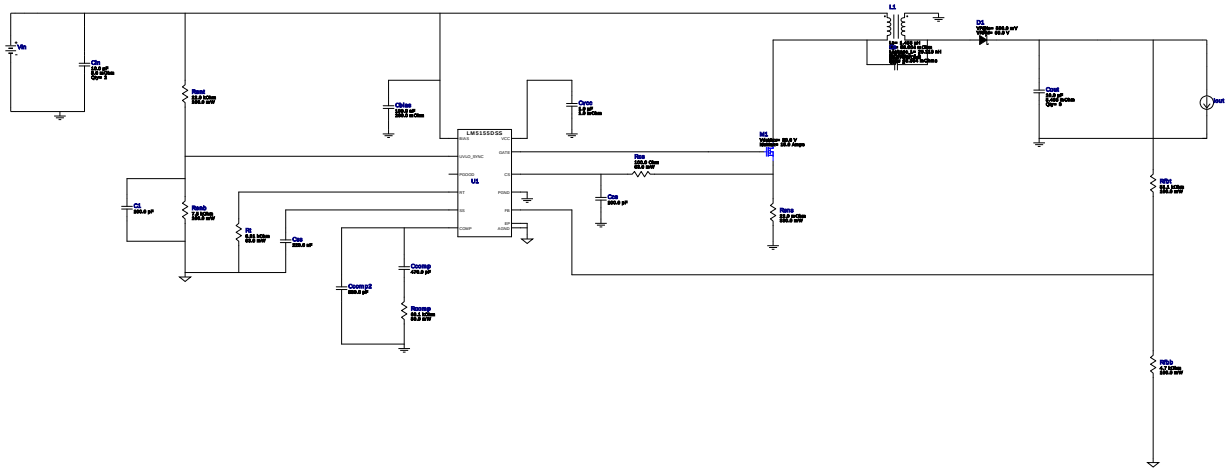
WEBENCH® Design Report

VinMin = 6.0V
VinMax = 18.0V
Vout = 12.0V
Iout = 1.0A

Device = LM5155DSSR
Topology = SEPIC
Created = 2026-10-13 07:08:03.981
BOM Cost = NA
BOM Count = 26
Total Pd = 1.41W

Design : 5 LM5155DSSR
LM5155DSSR 6V-18V to 12.00V @ 1A

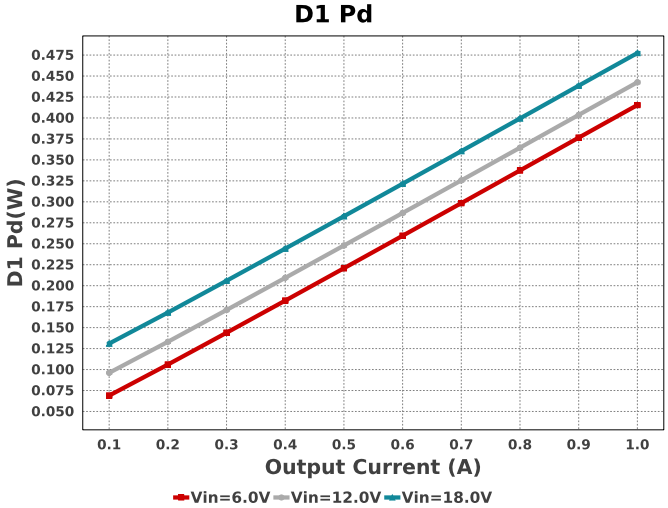


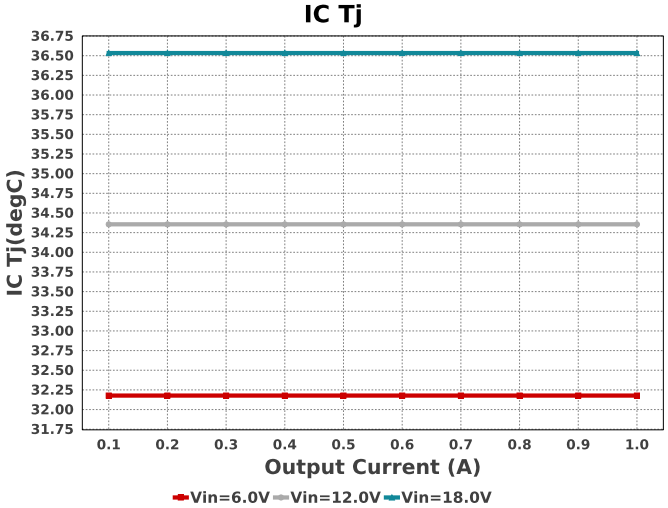
Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
C1	Kemet	C0201C101K3GACTU Series= C0G/NP0	Cap= 100.0 pF VDC= 10.0 V IRMS= 0.0 A	1	$0.01	0201 2 mm²
Cbias	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Ccomp	Yageo	CC0805JRNPO9BN471 Series= C0G/NP0	Cap= 470.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.02	0805 7 mm²
Ccomp2	Taiyo Yuden	UMK105CG331JV-F Series= C0G/NP0	Cap= 330.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Ccs	Samsung Electro-Mechanics	CL21C101JBANNNC Series= C0G/NP0	Cap= 100.0 pF VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Cin	Samsung Electro-Mechanics	CL32A106KBJNNNE Series= X5R	Cap= 10.0 uF ESR= 5.0 mOhm VDC= 50.0 V IRMS= 0.0 A	2	$0.15	1210_270 15 mm²
Cout	Taiyo Yuden	MSAST21GBB5106MTNA01 Series= X5R	Cap= 10.0 uF ESR= 5.463 mOhm VDC= 25.0 V IRMS= 2.36768 A	3	$0.04	0805 7 mm²
Cs	Taiyo Yuden	MSASG21GSB7105KTNA01 Series= X7R	Cap= 1.0 uF ESR= 25.874 mOhm VDC= 35.0 V IRMS= 1.56477 A	2	$0.04	0805 7 mm²
Css	TDK	C4532C0G1H224K320KA Series= C0G/NP0	Cap= 220.0 nF VDC= 50.0 V IRMS= 0.0 A	1	$0.58	1812 23 mm²
Cvcc	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²

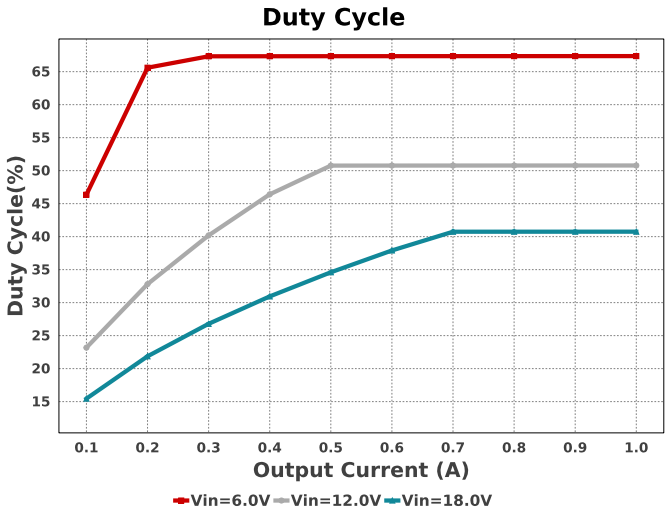
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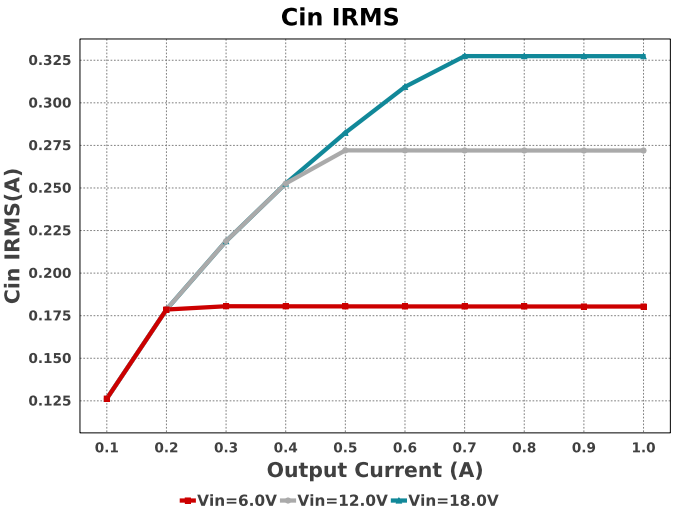
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
D1	Diodes Inc.	PDS760-13	VF@Io= 560.0 mV VRRM= 60.0 V	1	$0.36	PowerDI5 50 mm²
L1	CUSTOM	CUSTOM	Lp= 1.456 µH Rp= 58.804 mOhm Leakage_L= 29.118 nH Ns1toNp= 1.0 Rs1= 58.804 mOhms	1	NA	CUSTOM 0 mm²
M1	Texas Instruments	CSD87381P	VdsMax= 30.0 V IdsMax= 15.0 Amps	1	$0.44	MPC0005A 14 mm²
Rcomp	Yageo	RC0201FR-0768K1L Series= ?	Res= 68.1 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	0201 2 mm²
Rcs	Vishay-Dale	CRCW0402100RFKED Series= CRCW..e3	Res= 100.0 Ohm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Renb	Yageo	RT0603BRD077K5L Series= ?	Res= 7.5 kOhm Power= 100.0 mW Tolerance= 0.1%	1	$0.04	0603 5 mm²
Rent	Susumu Co Ltd	RG1608P-223-B-T5 Series= RG1608	Res= 22.0 kOhm Power= 100.0 mW Tolerance= 0.1%	1	$0.04	0603 5 mm²
Rfbb	Susumu Co Ltd	RG1608P-472-B-T5 Series= RG1608	Res= 4.7 kOhm Power= 100.0 mW Tolerance= 0.1%	1	$0.04	0603 5 mm²
Rfbt	Susumu Co Ltd	RG1608P-5112-B-T5 Series= RG1608	Res= 51.1 kOhm Power= 100.0 mW Tolerance= 0.1%	1	$0.06	0603 5 mm²
Rsns	Panasonic	ERJ-L14KF22MU Series= ERJ-L14	Res= 22.0 mOhm Power= 330.0 mW Tolerance= 1.0%	1	$0.11	1210 15 mm²
Rt	Vishay-Dale	CRCW04029K31FKED Series= CRCW..e3	Res= 9.31 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
U1	Texas Instruments	LM5155DSSR	Switcher	1	$0.80	DSS0012B 12 mm²

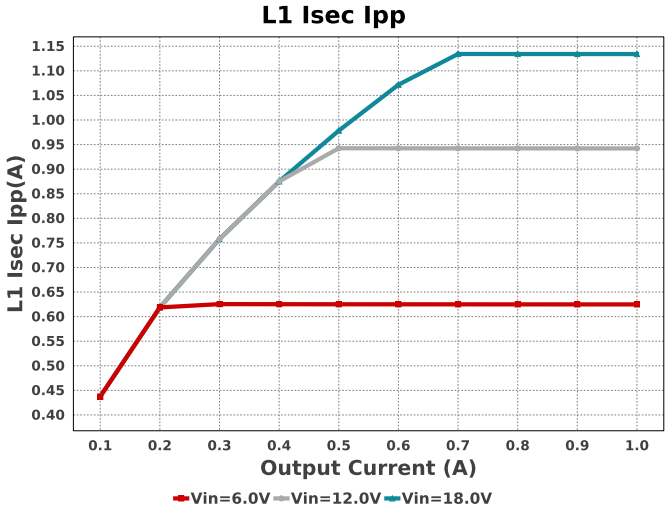


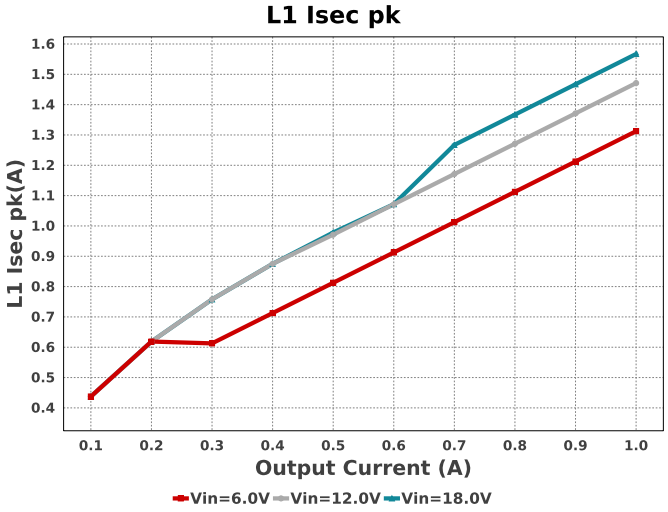


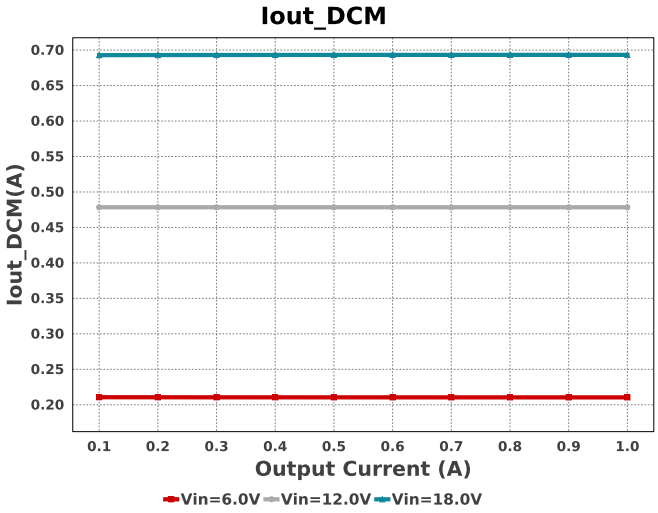
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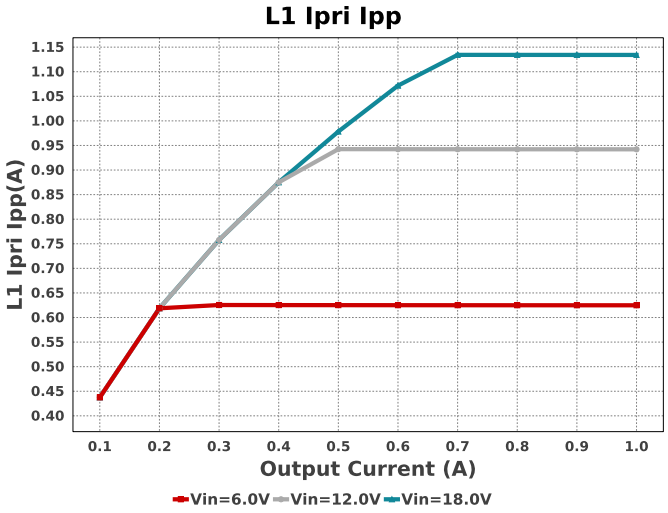




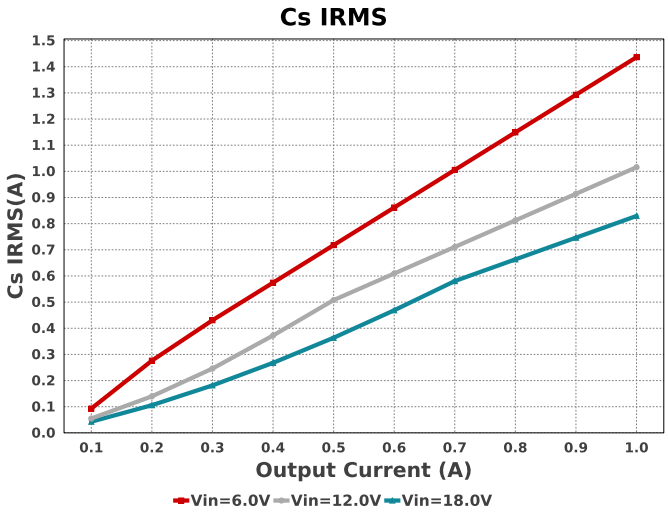


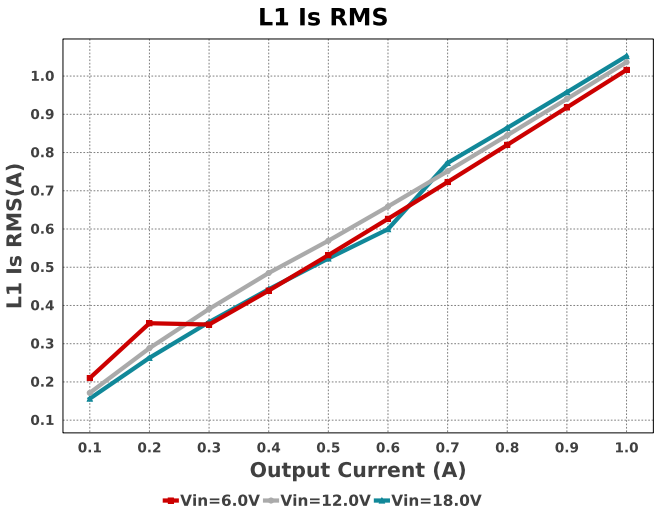


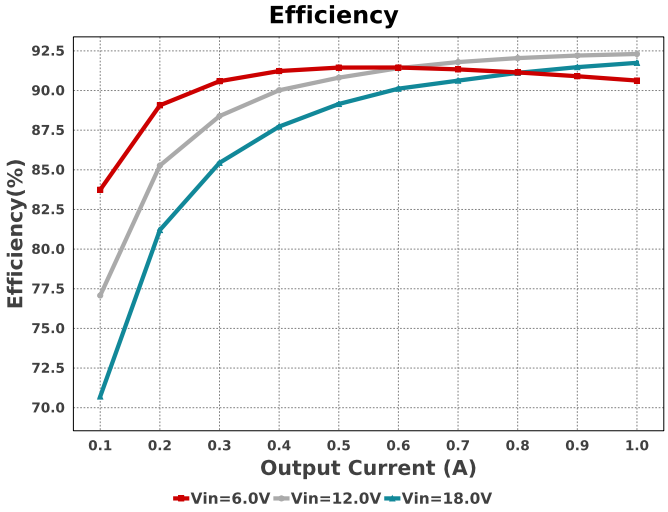


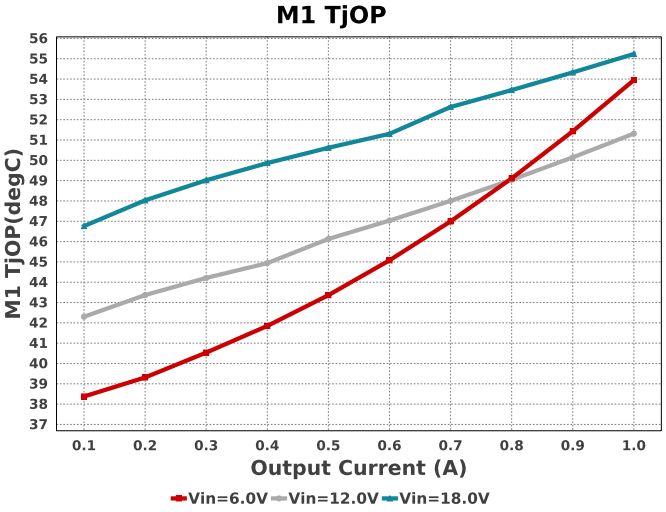


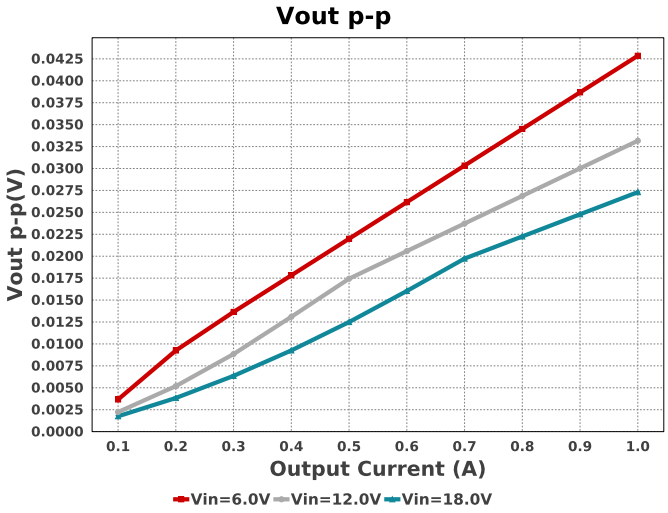
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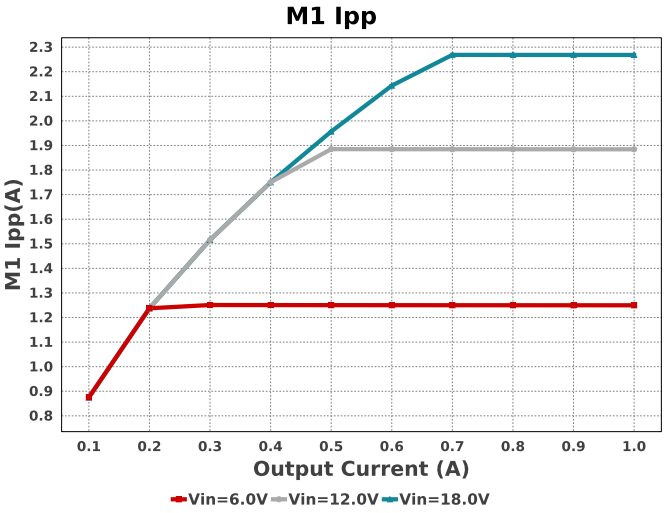




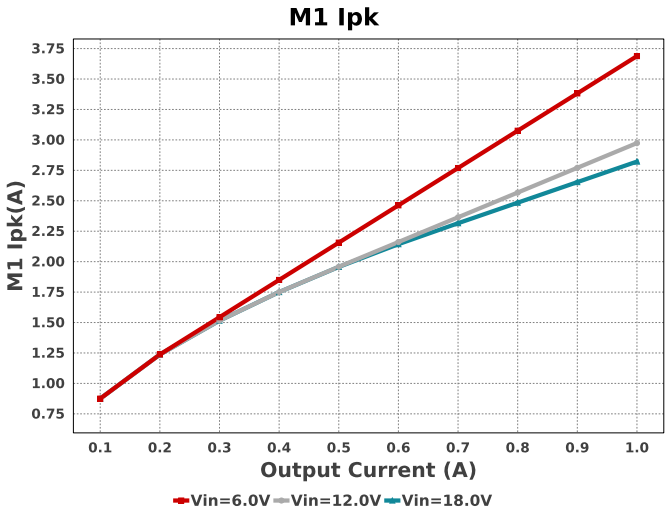


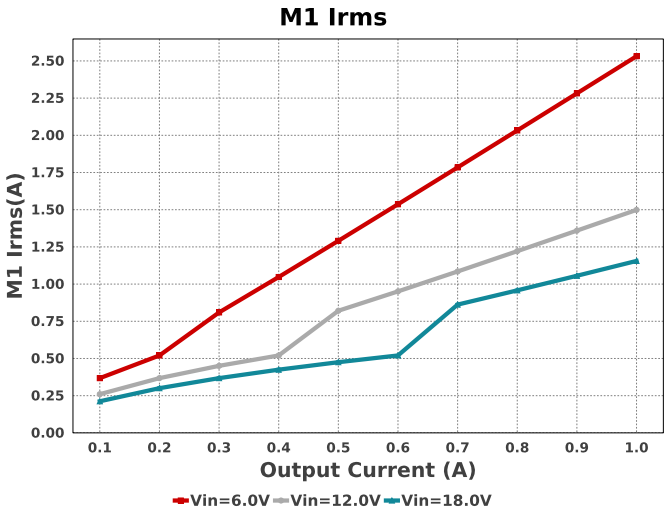


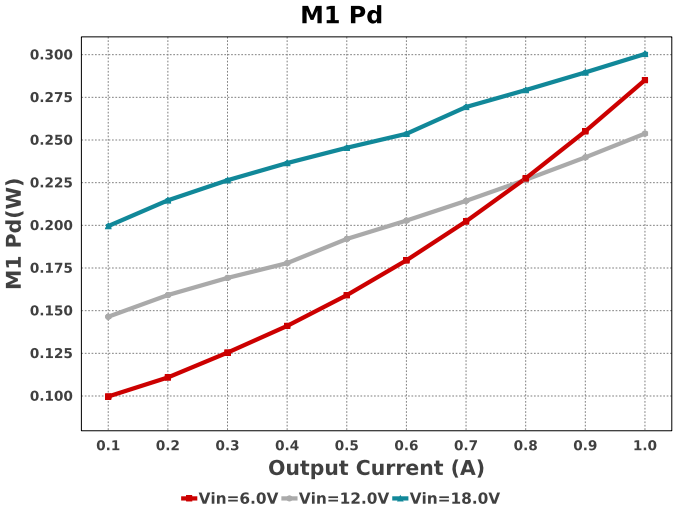


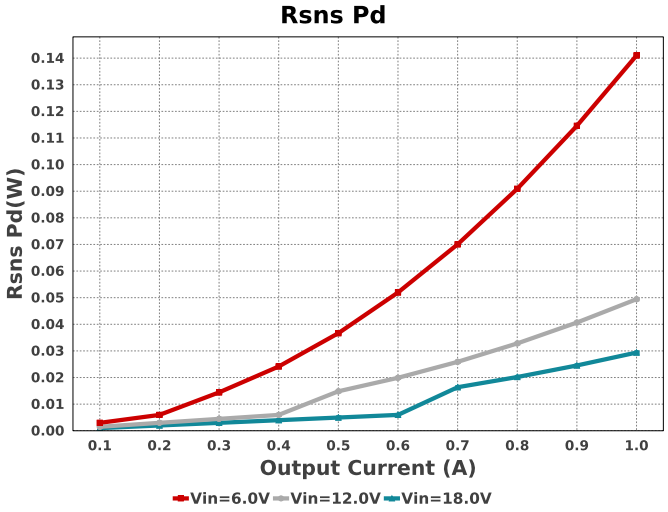


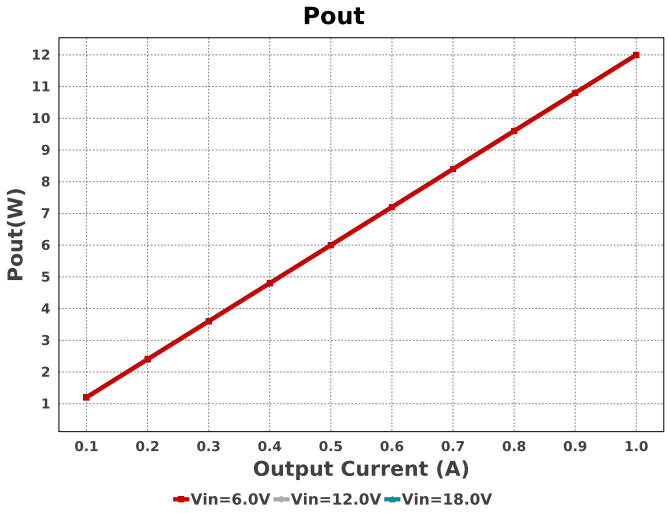
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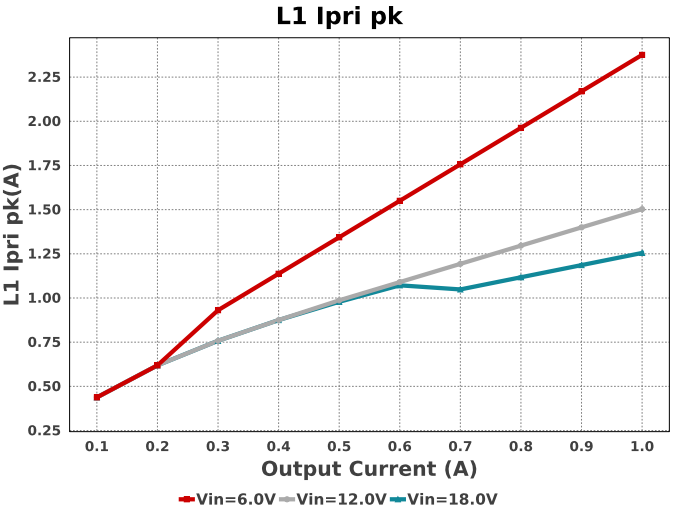




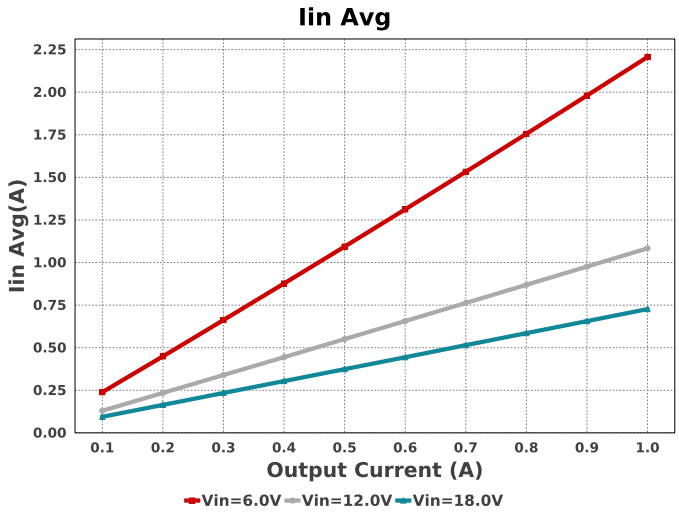


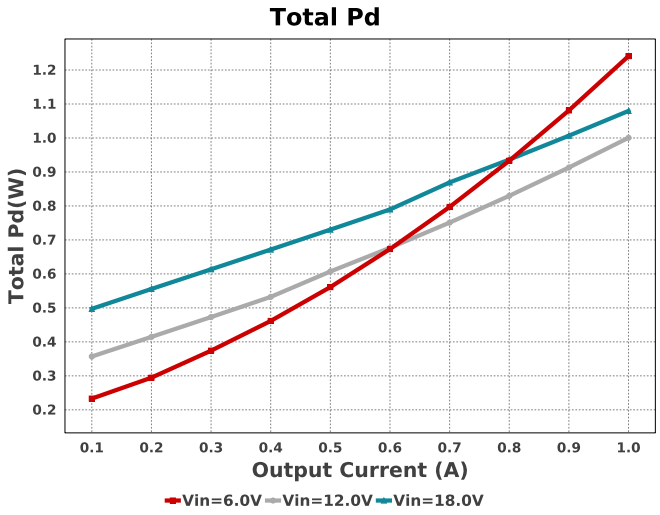


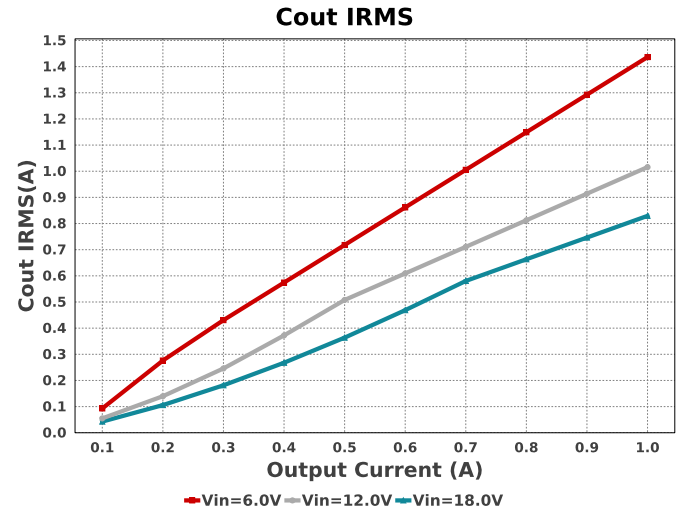


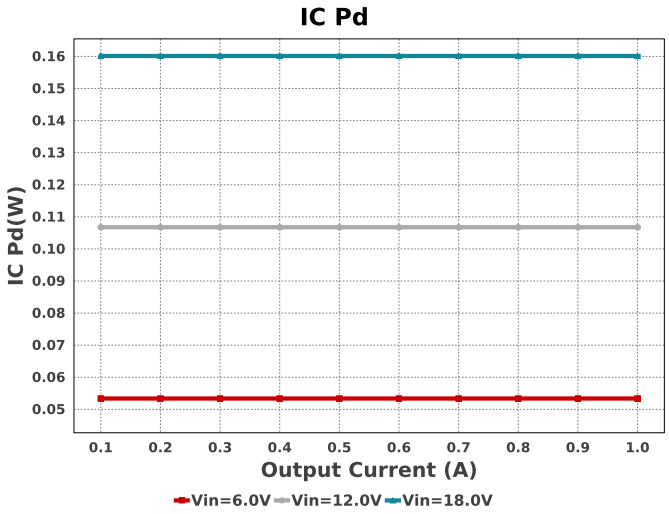


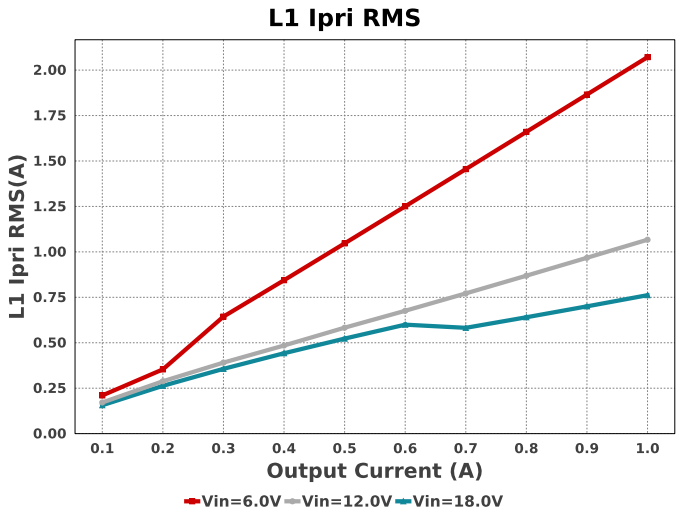
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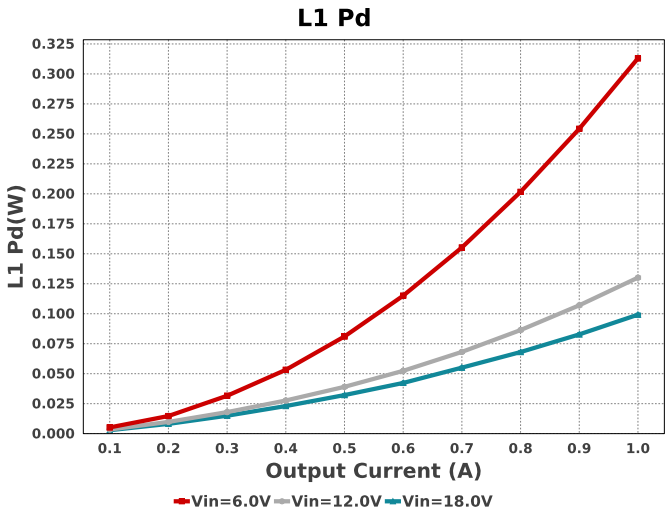












Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	180.38 mA	Capacitor	Input capacitor RMS ripple current
2.	Cout IRMS	1.436 A	Capacitor	Output capacitor RMS ripple current
3.	Cs IRMS	1.436 A	Capacitor	Coupling capacitor RMS ripple current
4.	M1 Ipk	3.688 A	Current	M1 peak current.
5.	D1 Pd	415.39 mW	Diode	Output Diode Power Dissipation
6.	IC Pd	117.97 mW	IC	IC power dissipation
7.	IC Tj	34.813 degC	IC	IC junction temperature
8.	ICThetaJA	40.8 degC/W	IC	IC junction-to-ambient thermal resistance
9.	Iin Avg	2.236 A	IC	Average input current
10.	M1 Irms	2.532 A	Mosfet	M1 MOSFET Irms
11.	M1 Pd	392.61 mW	Mosfet	M1 MOSFET total power dissipation

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#	Name	Value	Category	Description
12.	M1 TjOP	62.979 degC	Mosfet	M1 MOSFET junction temperature
13.	D1 Pd	415.39 mW	Power	Output Diode Power Dissipation
14.	IC Pd	117.97 mW	Power	IC power dissipation
15.	L1 Pd	313.0 mW	Power	Power Dissipation in the Inductor
16.	M1 Pd	392.61 mW	Power	M1 MOSFET total power dissipation
17.	Rsns Pd	140.99 mW	Power	Current Limit Sense Resistor Power Dissipation
18.	Total Pd	1.413 W	Power	Total Power Dissipation
19.	Rsns Pd	140.99 mW	Resistor	Current Limit Sense Resistor Power Dissipation
20.	BOM Count	26	System Information	Total Design BOM count
21.	Duty Cycle	67.357 %	System Information	Duty cycle
22.	Efficiency	89.465 %	System Information	Steady state efficiency
23.	FootPrint	433.0 mm^2	System Information	Total Foot Print Area of BOM components
24.	Frequency	2.153 MHz	System Information	Switching frequency
25.	Iout	1.0 A	System Information	Iout operating point
26.	Iout_DCM	210.44 mA	System Information	Approximate Current below which DCM mode of operation will begin
27.	M1 Ipp	1.25 A	System Information	M1 ripple pk-pk.
28.	Mode	CCM	System Information	Conduction Mode
29.	Pout	12.0 W	System Information	Total output power
30.	Total BOM	NA	System Information	Total BOM Cost
31.	Vin	6.0 V	System Information	Vin operating point
32.	Vout	12.0 V	System Information	Operational Output Voltage
33.	Vout Actual	11.872 V	System Information	Vout Actual calculated based on selected voltage divider resistors
34.	Vout Tolerance	2.788 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
35.	Vout p-p	42.852 mV	System Information	Peak-to-peak output ripple voltage
36.	L1 Ipri Ipp	624.853 mA	Transformer	L1 primary side ripple pk-pk.
37.	L1 Ipri RMS	2.071 A	Transformer	L1 primary rms current.
38.	L1 Ipri pk	2.376 A	Transformer	L1 primary pk current.
39.	L1 Is RMS	1.016 A	Transformer	L1 secondary rms current
40.	L1 Isec Ipp	624.853 mA	Transformer	L1 secondary side ripple pk-pk.
41.	L1 Isec pk	1.312 A	Transformer	Peak current in L1 secondary
42.	L1 Pd	313.0 mW	Transformer	Power Dissipation in the Inductor

Design Inputs

Name	Value	Description
Iout	1.0	Maximum Output Current
VinMax	18.0	Maximum input voltage
VinMin	6.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	LM5155	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

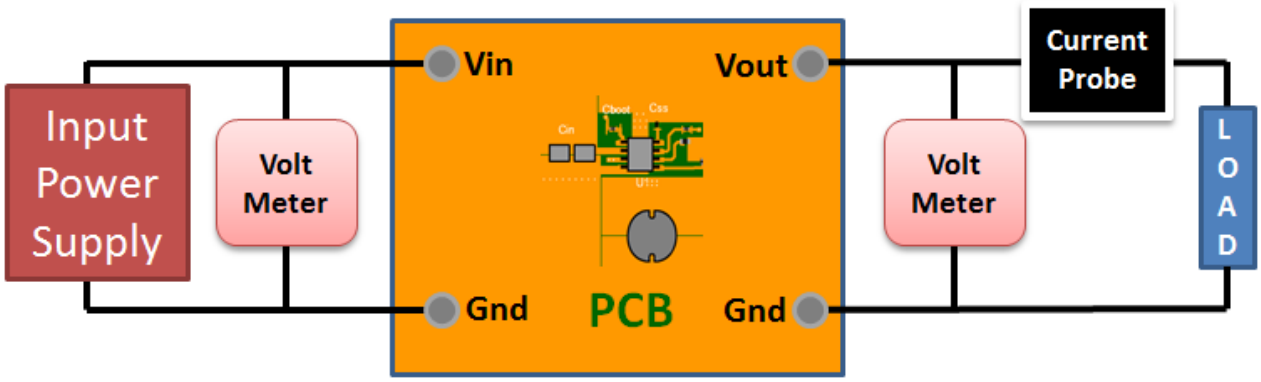
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 6.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : C0723DD09D79F8AFBAB58B2D8AAED5F5[v1]
2. **LM5155** Product Folder : http://www.ti.com/product/LM5155 : contains the data sheet and other resources.

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